

FAIRBANKS, AK, USA

WMO: 702610

Lat: 64.804N

Lon: 147.876W

Elev: 132

StdP: 99.75

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26411

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -41.2	(c) -38.6	(d) -37.1	(e) 0.1	(f) -33.4	(g) -36.0	(h) 0.1	(i) -32.8	(j) 7.9	(k) -14.2	(l) 6.5	(m) -15.2	(n) 0.2	(o) 20	(p) 0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	27.1	16.1	25.4	15.5	23.6	14.7	17.4	24.7	16.5	23.1	15.6	21.6	2.6	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.9	10.7	18.7	13.9	10.1	18.0	13.0	9.5	17.3	49.3	24.5	46.5	23.0	44.1	21.5	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.4	6.1	5.2	-42.5	30.2	2.6	1.7	-44.4	31.4	-45.9	32.4	-47.4	33.4	-49.3	34.6
			-36.1	19.2	1.1	1.4	-36.8	20.2	-37.5	21.0	-38.1	21.8	-38.9	22.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-22.3	-17.2	-11.8	1.0	10.2	16.0	17.1	13.9	7.9	-3.0	-15.5	-20.3	(6)
(7)		DBStd	15.68	9.99	8.33	7.36	5.93	4.37	3.16	2.75	3.28	3.99	6.27	7.73	8.33	(7)
(8)		HDD10.0	4954	1001	762	675	270	49	1	0	4	83	403	766	939	(8)
(9)		HDD18.3	7426	1259	995	933	519	252	82	56	143	312	661	1016	1197	(9)
(10)		CDD10.0	608	0	0	0	1	57	182	220	125	22	1	0	0	(10)
(11)		CDD18.3	37	0	0	0	0	1	13	17	6	0	0	0	0	(11)
(12)		CDH23.3	424	0	0	0	0	32	158	173	61	1	0	0	0	(12)
(13)		CDH26.7	66	0	0	0	0	3	28	24	9	0	0	0	0	(13)
(14)	Wind	WSAvg	1.8	0.9	1.2	1.9	2.5	2.6	2.4	2.1	2.0	1.9	1.6	1.1	0.8	(14)
(15)	Precipitation	PrecAvg	335	16	13	9	9	17	46	72	58	40	18	18	18	(15)
(16)		PrecMax	512	72	42	60	74	54	84	150	93	83	36	49	79	(16)
(17)		PrecMin	230	3	3	0	0	4	13	3	9	7	6	2	0	(17)
(18)		PrecStd	63	16	10	12	14	12	19	36	22	18	8	15	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.9	2.3	8.4	18.9	26.6	29.6	29.3	27.8	21.3	14.3	2.2	1.2	(19)
(20)			MCWB	-0.1	-0.2	2.7	9.3	13.2	16.5	17.4	17.5	12.5	8.2	0.3	-0.5	(20)
(21)		2%	DB	-2.4	-0.2	4.9	15.2	23.6	27.1	26.9	25.0	18.8	9.7	0.1	-2.6	(21)
(22)			MCWB	-3.4	-1.4	0.8	6.9	12.3	15.5	16.6	16.6	11.3	5.4	-0.7	-3.6	(22)
(23)		5%	DB	-5.2	-3.0	2.3	12.6	21.1	25.1	25.3	22.2	16.9	6.9	-2.2	-5.4	(23)
(24)			MCWB	-5.8	-4.1	-0.7	5.6	11.2	14.7	16.0	14.9	10.6	4.1	-3.0	-6.0	(24)
(25)		10%	DB	-8.3	-5.7	-0.4	10.3	18.8	23.2	23.6	20.0	14.9	4.6	-4.8	-8.2	(25)
(26)			MCWB	-8.8	-6.5	-2.8	4.4	10.0	14.0	15.4	14.0	9.8	2.5	-5.4	-8.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.6	0.6	3.1	9.6	14.2	17.9	18.8	18.8	13.1	8.8	0.9	-0.4	(27)
(28)			MCDWB	2.3	1.6	7.6	18.0	23.6	27.1	25.9	25.8	19.3	13.0	1.8	0.9	(28)
(29)		2%	WB	-2.9	-1.5	1.3	7.5	12.9	16.6	17.6	17.0	11.9	5.8	-0.7	-3.3	(29)
(30)			MCDWB	-1.9	-0.1	4.5	14.5	22.0	24.6	24.9	23.1	17.3	8.9	0.1	-2.4	(30)
(31)		5%	WB	-5.3	-4.0	-0.7	5.8	11.8	15.5	16.8	15.8	11.1	4.3	-3.0	-5.8	(31)
(32)			MCDWB	-4.7	-2.9	2.2	12.1	19.8	23.0	23.5	20.7	15.6	6.8	-2.2	-5.3	(32)
(33)		10%	WB	-7.9	-6.3	-2.7	4.6	10.7	14.7	16.0	14.7	10.3	2.6	-5.4	-8.4	(33)
(34)			MCDWB	-7.3	-5.5	-0.3	10.2	17.6	21.7	22.1	18.8	14.2	4.5	-4.8	-7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.6	11.1	14.3	12.2	11.9	10.8	9.8	9.4	9.7	7.7	7.8	8.3	(35)
(36)			MCDBR	9.9	11.6	15.9	14.8	15.3	14.0	13.4	13.4	13.4	10.7	8.0	10.8	(36)
(37)			MCWBR	9.1	10.1	12.0	8.1	6.6	5.5	5.4	6.0	7.0	6.9	7.1	10.0	(37)
(38)		5% WB	MCDWR	10.0	11.1	14.7	14.2	13.5	12.1	11.7	11.3	11.6	9.9	7.8	10.9	(38)
(39)			MCWBR	9.1	9.7	11.1	7.9	5.9	5.1	5.1	5.4	6.3	6.5	7.0	10.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.171	0.247	0.252	0.308	0.345	0.381	0.381	0.400	0.310	0.263	N/A	0.177	(40)	
(41)		taud	1.943	2.204	2.306	2.291	2.426	2.336	2.362	2.294	2.540	2.459	N/A	1.943	(41)	
(42)		Ebn at Noon	388	670	852	866	864	831	820	756	784	669	N/A	118	(42)	
(43)		Edh at Noon	23	54	76	100	97	109	104	101	64	42	N/A	8	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.16	0.85	2.65	4.58	5.26	5.54	4.79	3.66	2.30	0.94	0.24	0.05	(44)	
(45)		RadStd	0.01	0.07	0.17	0.29	0.39	0.42	0.33	0.36	0.28	0.10	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	+0.72	N/A	N/A	N/A	N/A	N/A	-270	-250	N/A	N/A	(46)
(47)	Regional (5 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	-326	-354	N/A	N/A	(47)

Nomenclature: See separate page